

CONDUCTIVITIES OF SOLID POLYMER ELECTROLYTE COMPLEXES OF ALKALI SALTS WITH POLYMERS OF METHOXYPOLYETHYLENEGLYCOL METHACRYLATES

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The ac conductivities of solid polyelectrolyte complexes of alkali salts (NaSCN , NaCF_3SO_3 , LiCF_3SO_3) with polymers derived from methoxypolyethyleneglycol methacrylates of the general formula $\text{CH}_2 = \text{C}(\text{CH}_3)\text{COO}(\text{CH}_2\text{CH}_2\text{O})_n\text{CH}_3$ ($n = 8, 22$) were measured as a function of temperature and salt content. The salts form crystalline or amorphous complexes with the polymers, some with a $\text{CH}_2\text{CH}_2\text{O}$ to cation ratio as low as unity. The results were compared with the thermal characteristics of the complexes obtained from differential scanning calorimeter measurements. Crystallinity is reduced and conductance enhanced by lowering the salt content, adding an external plasticizer like propylene carbonate or by incorporating a comonomer like styrene into the polymer. The Arrhenius conductance plots are linear, and the conductivities with those reported for alkali salt complexes with poly(ethylene oxide).

1. Introduction

Solvent-free solid polymeric electrolytes are under active investigation because of their potential application in high energy storage devices [1–8]. Most reports deal with the thermal and conducting characteristics of alkali salt complexes with poly(ethylene oxide), PEO, which in some instances have been cross-linked by radiation [6] or by means of urethane linkages [8]. PEO and its low molecular weight glyme analogues are known to form crystalline complexes with salts such as sodium thiocyanate, sodium iodide or sodium tetraphenylboron [1–9].

In recent years we extensively explored the cation-binding properties of linear and network polymers endowed with macrocyclic or linear polyethers (crown ethers and glymes) anchored to a polystyrene or polyacrylate backbone [10–14]. One of the studies concerned the optical properties and structures of alkali picrates solubilized in cross-linked, transparent films consisting of a copolymer of methylmethacrylate and the cation-binding monomer methoxypolyethyleneglycol methacrylate [14]. The salt-solubilizing properties of these materials prompted us to look at the thermal and conducting properties of solid state mixtures

of lithium and sodium salts with the homopolymers of methoxypolyethyleneglycol methacrylate. The results suggest that these systems may be viable alternatives for conducting solid polyelectrolyte complexes.

2. Experimental

2.1. Materials

Two methoxypolyethyleneglycol methacrylates of the general formula $\text{CH}_2 = \text{C}(\text{CH}_3)\text{COO}(\text{CH}_2\text{CH}_2\text{O})_n\text{CH}_3$ ($n = 8$ and 22) were polymerized in benzene at 65°C with azobisisobutyronitrile as initiator. Details of the polymerization have been reported elsewhere [14]. The number average molecular weights of the polymers as measured by membrane osmometry ranged from 150000 to 300000. The two polymers have been abbreviated in this discussion as PMG8 and PMG22. Solid complexes with alkali salts were obtained by slow evaporation of a homogeneous polymer salt solution (usually in absolute methanol), or directly from the melt. The complexes were dried under vacuum for several days.

2.2. Measurements

Sample preparation and the design of the conductance cell basically follow procedures outlined by Wright [15]. Pellets (1–2 mm thick, about 1 cm in diameter) of carefully dried polymer-salt complexes and sandwiched in between two thin copper plates were made in a die by applying a pressure of 10 tons. The pellet was then placed in an thermostated conductance cell [15] and the ac conductivity measured under a stream of dry helium by means of a 1673 General Radio Corporation automatic capacitance bridge coupled with a 1672 digital control unit. A thermocouple close to the pellet registered the temperature. The thermal behavior of the complexes was studied with a Perkin Elmer DSC-4 differential scanning calorimeter using a heating rate of 20°C per minute [14]. The heating-cooling sequence was repeated until reproducible scans were obtained (in most instances the second scan). The glass transition temperature T_g was recorded as the inflection point and the melting point T_m as the maximum of the melt peak.

3. Results and discussion

Polymer PMG8 is a viscous melt with $T_g = -58^\circ\text{C}$, while PMG22 is a white crystalline solid melting at 44°C . Sodium thiocyanate and PMG22 form a crystalline complex with $T_m = 139^\circ\text{C}$ (fig. 1), and at a

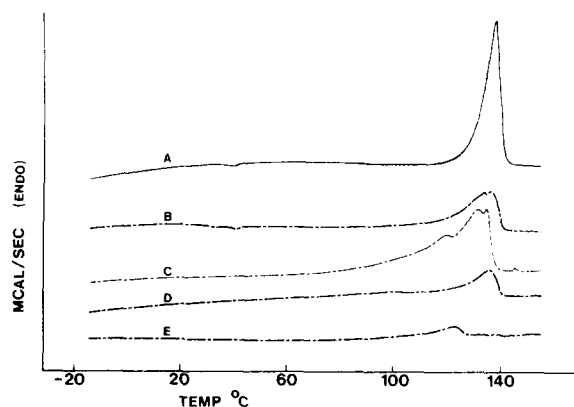


Fig. 1. DSC curves for the complex of PMG22 with NaSCN ($\text{CH}_2\text{CH}_2\text{O}/\text{Na} = 3$). Systems B, C, D and E contain 30%, 40%, 50% and 60% (by weight) of propylene carbonate, respectively, while A has no plasticizer added.

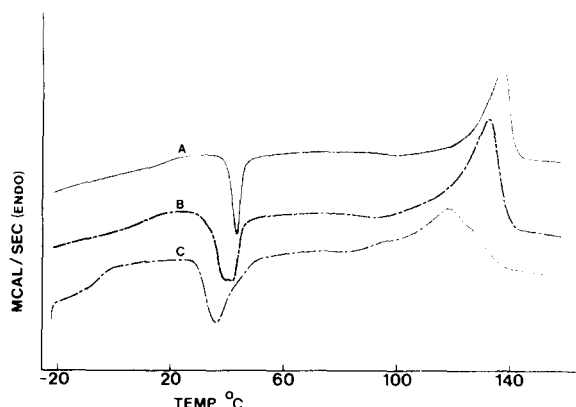


Fig. 2. DSC curves for the PMG22/NaSCN complex at ratios $\text{CH}_2\text{CH}_2\text{O}/\text{Na} = 4$ (A), 5 (B) and 6 (C).

ratio $\text{CH}_2\text{CH}_2\text{O}/\text{Na} = 3$ no free NaSCN ($T_m = 300^\circ\text{C}$) is found in the DSC scan [14]. Free salt is detected at higher salt content, i.e., at $\text{CH}_2\text{CH}_2\text{O}/\text{Na}$ ratios of 1 or 2. At ratios of 4, 5 or 6 the thermal behavior differs markedly from that of the 3:1 mixture. Fig. 2 shows that on cooling the melt below -20°C , the DSC scan on subsequent heating first depicts a glass transition (between -20 and 20°C , depending on salt content), then a crystallization peak at about 40°C (close to the T_m of free PMG22), and eventually the melt peak around 140°C . The melt peak is more diffuse and its maximum shifts to lower temperature when the salt content is decreased. The crystallization peak at 40°C is not found for the 3:1 mixture (fig. 1). Apparently, the crystalline complex requires three $\text{CH}_2\text{CH}_2\text{O}$ moieties per sodium ion, and that only for a 3:1 mixture of PMG22 and NaSCN does crystallization occur without delay on rapid cooling. At ratios of $\text{CH}_2\text{CH}_2\text{O}$ to Na above 3, the crystallinity is reduced although it is not certain if the complex itself retains three $\text{CH}_2\text{CH}_2\text{O}$ units per sodium ion.

Arrhenius-type conductance plots of some PMG22/NaSCN mixtures are depicted in fig. 3. The high melting 3:1 crystalline complex, as expected, has a low conductance, but a decrease in salt content and the accompanying loss in crystallinity rapidly enhances the conductance. Its value is about three orders of magnitude higher for the 6:1 mixture than for the 3:1 complex. Those for the 4:1 and 5:1 mixtures fall in between. The behavior of these salt complexes with the glyme side chains of the polymer resembles that of the PEO/Na complexes, except that in these latter systems the

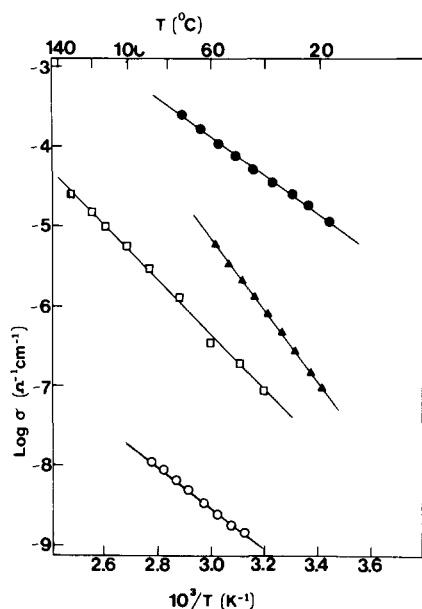


Fig. 3. Arrhenius conductivity plots for the systems PMG22/NaSCN with $\text{CH}_2\text{CH}_2\text{O}/\text{Na} = 3$ ($\circ$, $\bullet$, the latter with 50% propylene carbonate added) and $\text{CH}_2\text{CH}_2\text{O}/\text{Na} = 6$ ($\square$), and for PMG22/ NaCF_3SO_3 , $\text{CH}_2\text{CH}_2\text{O}/\text{Na} = 3$ ($\blacktriangle$).

sodium ion appears to be tetra-coordinated [5]. It is generally believed that the cation is the mobile species, and its transport through crystalline and amorphous regions of the polymer has been thoroughly discussed [5]. It is not clear why the anion should be immobile except for its larger size, since its interaction with the polyether chain is weak in comparison to alkali ions.

Low salt content reduces crystallinity and makes empty oxygen sites available for cation transport. Reduction of crystallinity can also be accomplished by adding an external plasticizer. Fig. 1 shows that crystallinity of the 3:1 PMG22/NaSCN mixture is greatly reduced on adding propylene carbonate. This in turn also raises the conductance. At 60°C the addition of 50% by weight of propylene carbonate to the 3:1 complex of PMG22/NaSCN enhances the conductance from 3×10^{-9} to more than $10^{-4} \Omega^{-1} \text{cm}^{-1}$. However, the material becomes softer. This also occurs when the salt content is lowered, making it more difficult to measure conductances at elevated temperatures.

Another system of interest is that of sodium triflate and PMG22. DSC scans for a mixture with $\text{CH}_2\text{CH}_2\text{O}/\text{Na}$ equal to unity reveal a homogeneous crystalline

complex melting at $T_m = 317^\circ\text{C}$ and no detectable peak at 252°C , the melt peak for free NaCF_3SO_3 [14]. Hence, the complex has a higher melting point than the free salt, and apparently contains one $\text{CH}_2\text{CH}_2\text{O}$ per sodium ion! The product in the molten state is transparent. At $\text{CH}_2\text{CH}_2\text{O}/\text{NaCF}_3\text{SO}_3$ ratios higher than unity the melt peaks become progressively more diffuse. The lower crystallinity and empty oxygen sites again provide better transport for the complexed ions. This is indicated by comparing the conductances of the 3:1 mixtures of PMG22 with NaSCN and NaCF_3SO_3 . The latter is two orders of magnitude larger than for NaSCN (fig. 3).

Crystalline complexes are more difficult to form when the glyme side chains are short as in PMG8. No melt-peaks are found with complexes of PMG8 and NaSCN. The homogeneous mixtures only show a shift in T_g to higher temperatures with increasing salt content. This behavior was also reported for lithium salt complexes with PMG8 [14]. For example, amorphous, transparent materials are formed when LiCF_3SO_3 is dissolved in PMG8, and the systems remain homogeneous down to a $\text{CH}_2\text{CH}_2\text{O}/\text{Li}$ ratio of unity. Such a high salt content raises the T_g from -58°C for pure PMG8 to 71°C for the 1:1 complex. For the 3:1 mixture ($T_g = 27^\circ\text{C}$) the conductance at 60°C was found to be $10^{-4} \Omega^{-1} \text{cm}^{-1}$. Crystalline complexes are formed between PMG22 and lithium salts, e.g., a sharp meltpeak at 141°C is found with lithium chloride, but generally the crystallization with lithium salts appears to be more difficult than with sodium salts. For example, LiCF_3SO_3 forms crystalline complexes with PMG22 only when cooling is slow, and the melt peaks are more diffuse than with NaCF_3SO_3 .

Monomers such as styrene may be used to form copolymers with MG22. The noncomplexing styrene units may then function as a non-volatile internal plasticizer to improve mechanical properties and enhance the conductance by lowering the crystallinity of the polyelectrolyte complex. Fig. 4 shows the Arrhenius conductance plots for complexes of LiCF_3SO_3 with PMG22 and with a 6:1 molar ratio copolymer of styrene and MG22 (in both systems the $\text{CH}_2\text{CH}_2\text{O}/\text{Li}$ ratio was five). The copolymer complex exhibits conductances that at comparable temperatures are at least an order of magnitude higher than for the homopolymer. For the copolymer

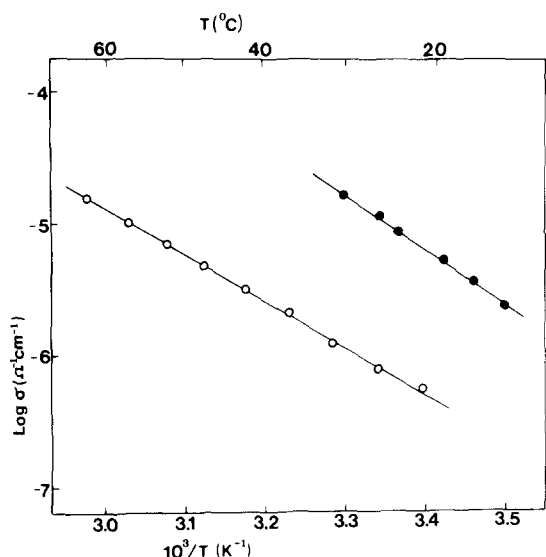


Fig. 4. Arrhenius conductivity plots for PMG22/LiCF₃SO₃ (○) and for the copolymer (PMG22)(Polystyrene)₆ with LiCF₃SO₃ (●). In both systems CH₂CH₂O/Li = 5.

the conductance is already $10^{-5} \Omega^{-1} \text{ cm}^{-1}$ at room temperature, while this value for PMG22 is reached at 60°C. A complex of PEO with LiCF₃SO₃ (CH₂CH₂O/Li = 5) was reported to have a conductance of $10^{-5} \Omega^{-1} \text{ cm}^{-1}$ at 100°C [16].

The examples presented here suggest that the alkali salt complexes with oligo-oxyethylene side chain polymers may be viable alternatives for the poly(ethylene oxide) complexes. Polymer backbones such as polystyrene or poly(phenylene oxide) would probably be better suited for electrochemical applications than a polyacrylate backbone. However, thermal and conducting properties are mainly determined by the length of the side chain. Other systems are presently being looked into, e.g., complexes with LiClO₄ and other salts. Mechanical properties are a problem because the materials become softer when the salt content is lowered. However, some experiments with radiation crosslinking show improved mechanical properties for the polymer electrolyte complexes without significantly altering the conducting characteristics. We are also exploring ways to chemically crosslink the polymers, and to investigate block copolymers containing comonomer units with a high glass transition temperature.

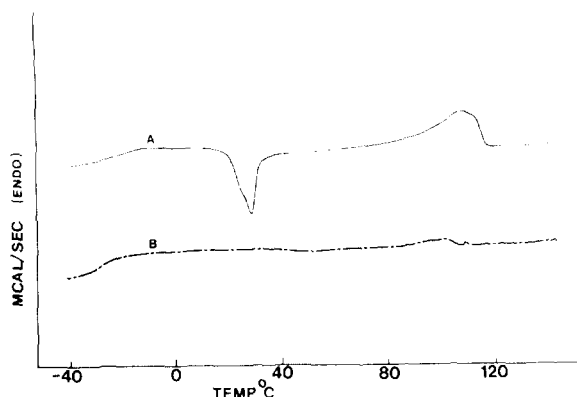


Fig. 5. DSC curves for PMG22/LiCF₃SO₃ (A) and (PMG22)(Polystyrene)₆/LiCF₃SO₃ (B), ratio CH₂CH₂O/Li = 5 in both systems.

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